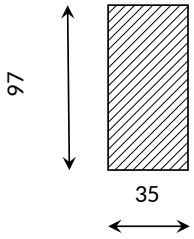
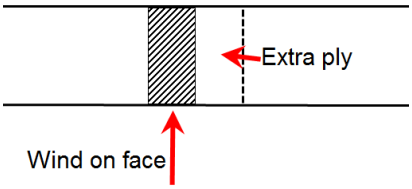
**Plated spandrel panel stud design check to EC5 (Flatwise timber):**



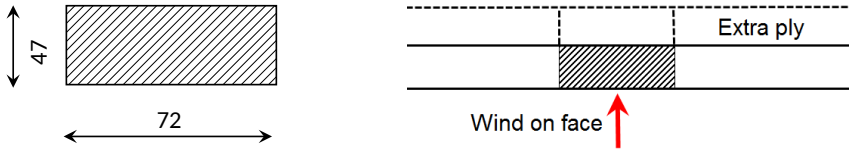

| | | | | | |
|---|---|---|---|---|---|
| Roof pitch: | | 30 deg | | | |
| Roof load: | dead | 0.685 kN/m² | Site snow: | 0.480 | kN/m² |
| Gable ladder overhang: | | 0.600 m | Distance to 1st truss | 0.050 | m |
| Dynamic wind pressure: | | 0.60 kN/m² | | | |
| Wall type | | | | | |
| | | self wt | 0.100 | kN/m² | |
| max stud length | | 3.400 m | stud c/c | 0.60 | m |
| Timber | 35 mm (Width) | 97 mm (Depth) | | 1 | ply |

Check gable panel for wind on face as leading action combined with vertical loads
PL NA.1: short-term wind action; confirm external and internal wind coefficients.
OK/FAIL: kontrola slupka dla wpisanych zalozen; polaczenia i kombinacje wymagaja osobnej weryfikacji.

| | | | | |
|---|---|---|---|---|
| Design axial stress | 0.254 N/mm² | | Design bending stress | 16.349 N/mm² |
| Design axial strength | 0.432 N/mm² | | Design bending strength | 18.129 N/mm² |
| kc,min | 0.030 | | | |
| csi | 1.490 | FAIL | | |
| deflection | 24.91 mm | allowable deflection | 15.3 mm | FAIL |

Deflection limit is a project assumption, not a verified PL limit for studs.

**Plated spandrel panel stud design check to EC5 (Edgewise timber):**




| | | | | | |
|---|---|---|---|---|---|
| Roof pitch: | | 30 | deg | | |
| Roof load: | dead | 0.685 | kN/m² | Site snow: | 0.480 kN/m² |
| Gable ladder overhang: | | 0.600 | m | Distance to 1st truss | 0.050 m |
| Dynamic wind pressure: | | 0.600 | kN/m² | | |
| Wall type: | | | | | |
| | | self wt | | 0.100 kN/m² | |
| max stud length | | 3.60 | m | stud c/c | 0.60 m |
| Multi-ply fixing | nail dia | 3.35 | mm | nail spacing | 300 mm |
| Timber | | 72 | mm | 47 mm | ply |
| | | Width | | Depth | |

Check gable panel for wind on as leading action face combined with vertical loads
PL NA.1: short-term wind action; confirm external and internal wind coefficients.
OK/FAIL: kontrola slupka dla wpisanych zalozen; nie potwierdza nosnosci lacznikow.

| | | | | | |
|---|---|---|---|---|---|
| Design axial stress | | 0.090 | N/mm² | Design bending stress | 9.720 N/mm² |
| Design axial strength | | 0.688 | N/mm² | Design bending strength | 20.956 N/mm² |
| kc,min | | 0.047 | | | |
| csi | | 1.967 | FAIL | | |
| deflection | | 26.77 | mm | allowable deflection | 16.2 mm FAIL |

Deflection limit is a project assumption, not a verified PL limit for studs.
Calculation of effective EI and resulting bending stress for two or three members nailed together
PN-EN 1995-1-1:2010 Annex B; fastener resistance must be verified separately.